



Identity, Reset, and Input

Team studio sheet - switch Driver and Navigator every challenge.

Team members: _____

Date: _____

Our numbered kit and device

Kit _____ Robot/device name _____ Driver first _____ Navigator first _____

Challenge 1 - prove identity and safe reset

Predict	Run	Evidence	Safe reset
Which prepared output will respond?	Trigger the mentor-approved test.	Write or sketch what proved this is our robot.	Output off / controller ready / roles switch

Challenge 2 - measure an input before choosing a threshold

Condition	Prediction	Actual sensor value	Same unit?	What we noticed
Far			Yes / fix	
Near			Yes / fix	
Far again			Yes / fix	

Our threshold and why

A value between our near and far readings is _____ because...

Team checkpoint

Both teammates can point to the real readings and explain why copying another team's threshold may not work.



Outputs and a Full Interaction

Use only mentor-marked ports, prepared connections, and safe servo angles.

Team members: _____

Date: _____

Challenge 3 - two different outputs

Output	Prediction	What we ran	What actually happened	Safe state
LED		0% -> 50% -> 100% -> 0%		0%
Servo		Only the two marked safe angles		Ready angle
Together		Predict, then run both		Both reset

Challenge 4 - input -> decision -> two outputs -> reset

Say the rule before running it

IF the object is _____ than _____, THEN the light _____ AND the pointer _____; OTHERWISE...

Trial	Object condition	Prediction	LED	Servo/pointer	Did reset work?
1	Far				
2	Near				
3	Far again				

Change one approved variable, then retest

We changed only _____. We held _____ the same. The result was...

Final proof

Both teammates can explain the input, threshold decision, two outputs, and what the system does when the condition is false.



Three Interactions Worth Discussing

Each idea needs an input, a decision, an output, and a reason people would care.

Team members: _____

Date: _____

Problem, delight, or future possibility

Who is the interaction for, and what should be easier, safer, clearer, or more interesting?

Sketch three different IF / THEN systems

1

It responds when...

2

It responds when...

3

It responds when...

Idea	Input	Decision / threshold	Output(s)	Why it matters
1				
2				
3				

Choose or combine

We want to explore idea(s) _____ because the interaction would...



Map the System and First Test

Build the smallest safe interaction that can answer one useful question.

Team members: _____

Date: _____

INPUT	DECISION	OUTPUT 1	OUTPUT 2	OTHERWISE / RESET
What changes in the world?	What condition or threshold?	What happens?	What else happens?	How does it return safely?

Three-frame interaction storyboard

BEFORE
Show person, sensor/input, and visible result.

TRIGGER
Show person, sensor/input, and visible result.

RESPONSE + RESET
Show person, sensor/input, and visible result.

First prototype question
Can we prove that _____ causes _____ safely and repeatably?

First build roles	Names / contribution
Hardware guardian	
Coder / logic explainer	
Test and evidence lead	

Mentor conversation - not a score
Circle together: system is explainable / needs another conversation; safe first test / make smaller; ready to prototype / repair plan first.